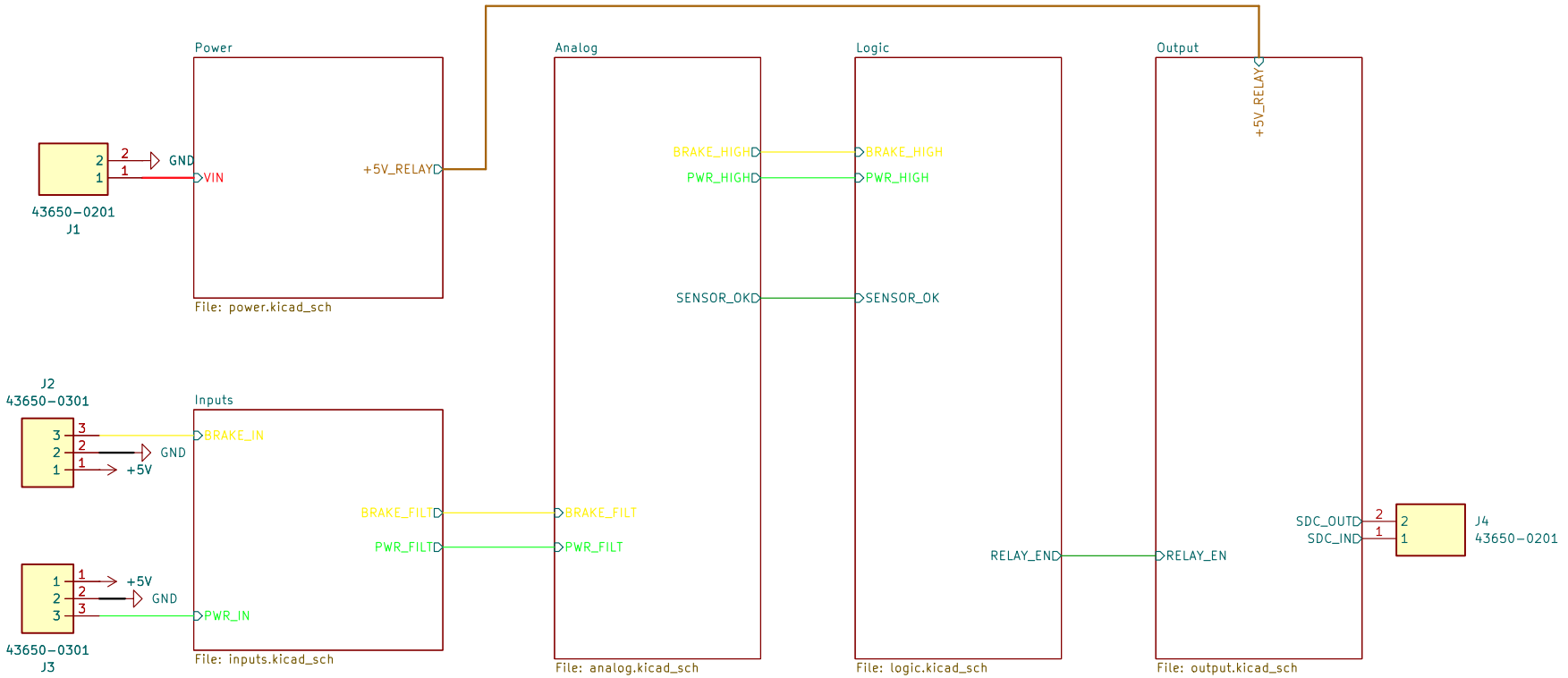
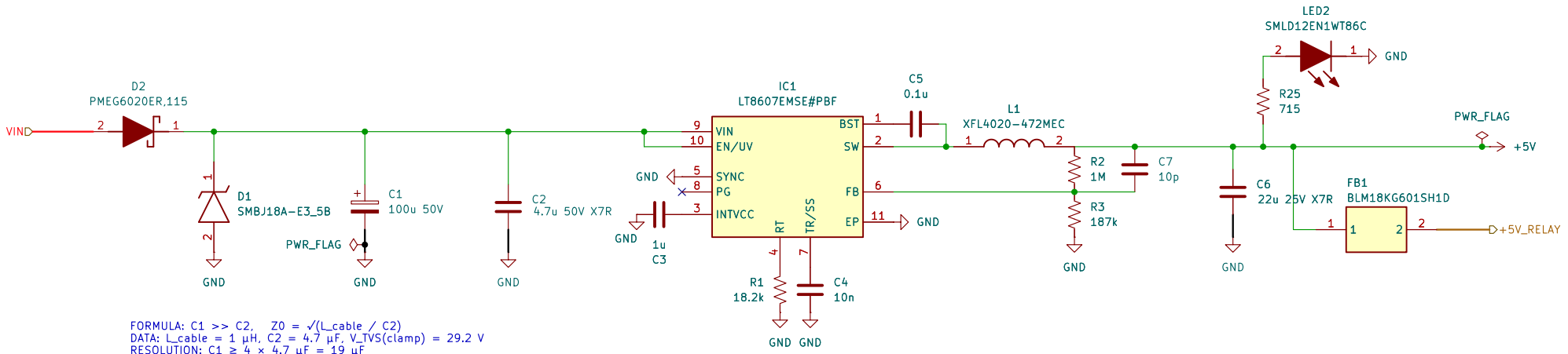


Massimo Maglio
E-Agle TRT

Sheet: recruitment
File: recruitment.kicad_sch

Title: Task 2 – Brake System Plausibility Device

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FORMULA: C1 >> C2, Z0 = √(L_cable / C2)
DATA: L_cable = 1 µH, C2 = 4.7 µF, V_TVS(clamp) = 29.2 V
RESOLUTION: C1 ≥ 4 × 4.7 µF = 19 µF
Z0 = sqrt(1 µH / 4.7 µF) ≈ 0.46 Ω

The chosen switching frequency is 2 MHz:
it is below the maximum frequency allowed by the buck converter and above the AM radio band.

The inductor was chosen according to the datasheet reference design (4.7 µH).

FORMULA:
ripple = V_OUT / (L × f_SW) × (1 − V_OUT / V_IN(MAX)) (datasheet p. 14)
Capacitor charging: I_C = C × V / t
I_PEAK = I_C + I_LOAD + ripple / 2

DATA:
V_OUT = 5 V, V_IN(MAX) = 17 V, L = 4.7 µH, f_SW = 2 MHz, I_LOAD = 45 mA,
C = 35 µF (total capacitance on the 5 V rail), t = 3.9 ms (soft-start time, datasheet p. 16)
RESOLUTION:
ripple = 5 V / (4.7 µH × 2 MHz) × (1 − 5 V / 17 V)
= 0.53 A × 0.71
≈ 0.38 A

I_C = 35 µF × 5 V / 3.9 ms
≈ 0.05 A

I_PEAK = 0.05 A + 0.045 A + 0.38 A / 2
= 0.05 A + 0.045 A + 0.19 A
≈ 0.30 A

The feedback resistors were chosen according to the datasheet reference design
(R2 = 1 MΩ, R3 = 187 kΩ).

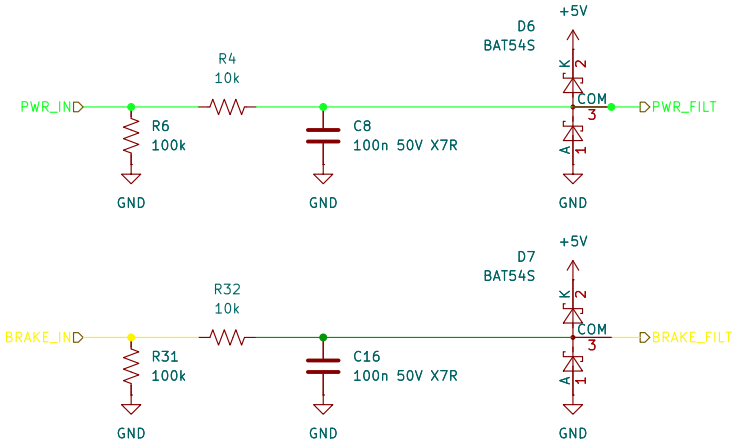
FORMULA:
V_OUT = V_FB × (R2 + R3) / R3 (datasheet p. 13)

DATA:
V_FB = 0.778 V (internal reference, datasheet)
R2 = 1 MΩ, R3 = 187 kΩ

RESOLUTION:
V_OUT = 0.778 V × (1 MΩ + 187 kΩ) / 187 kΩ ≈ 4.94 V

FORMULA: C_OUT = 100/(V_OUT/f_sw), ΔV_OUT ≈ ΔI_L / (8 × f_SW × C_OUT)
DATA: V_OUT = 5V, f_Sw = 2, ΔI_L = 0.38A(inductor ripple current)
RESOLUTION: C_OUT = 100 / (5 × 2)
C_OUT = 10 µF (at minimum)
ΔV_OUT ≈ 0,38 A / (8 × 2 MHz × 22 µF)
ΔV_OUT ≈ 1,1 mV

Sheet: Power
File: power.kicad_sch
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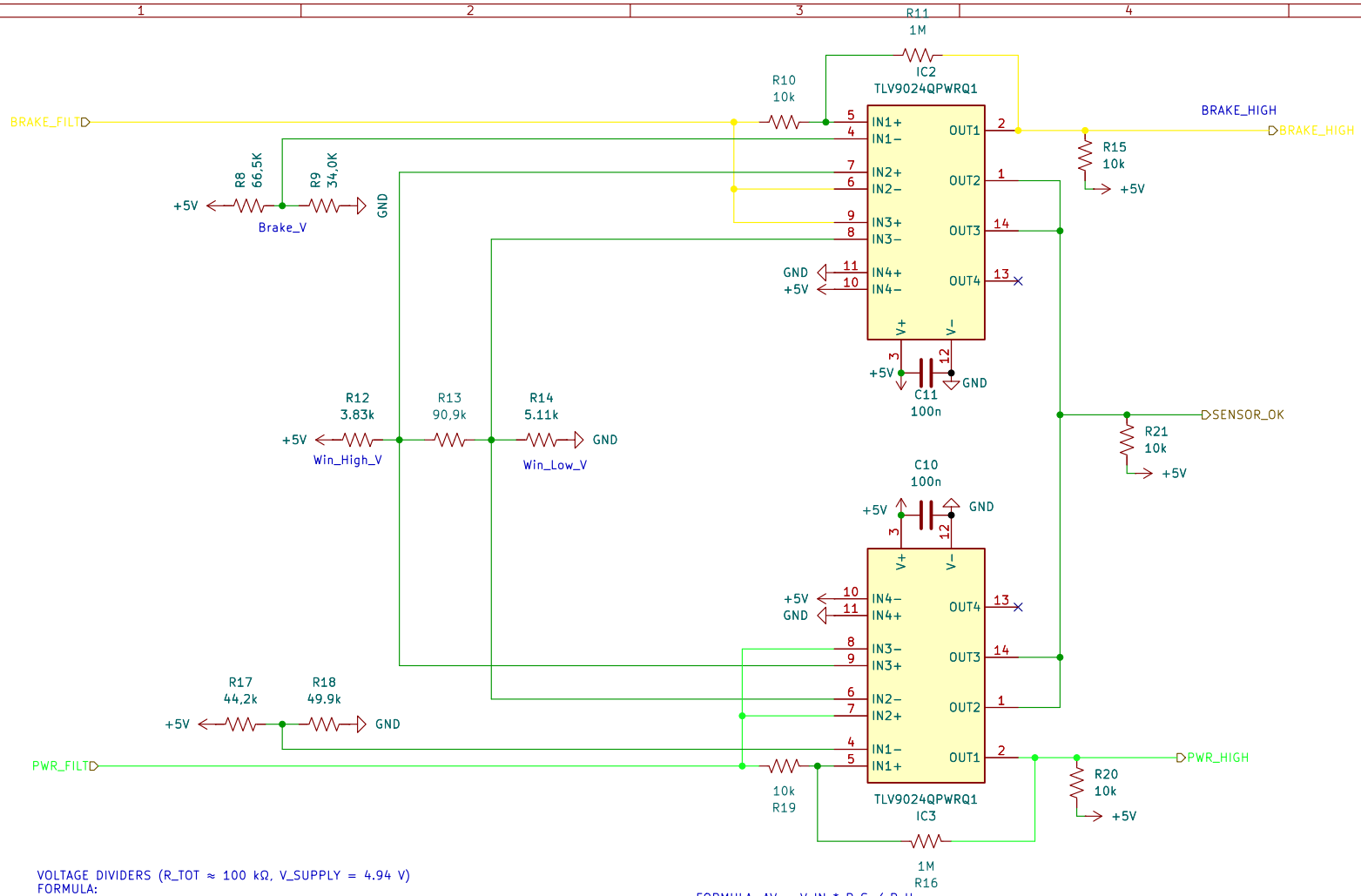


ASSUMPTION: Fastest pedal stomp: 50ms, Inverter's noise: 10kHz
Delay: τ = 1ms (5τ = 5ms, 1% of the 500ms)
FORMULAS: f = 0.35/t, f_C = 1/(2π*t)
RESOLUTION: f = 0.35/0.05s = 7Hz -> 10Hz
f_C = 1/(2π*1ms) = 160HZ(teorical)(2π*t)
C_F = 100nF(standard)
R_TOT = 1/(2π*160 Hz*100 nF) = 9.95kΩ
R_IN = 9.95k − 100 = 9.85 -> 10K
CONFIRMATION:
f_C = 1 / (2π × 10.1 kΩ × 100 nF) = 158HZ

Sheet: Inputs
File: inputs.kicad_sch

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VOLTAGE DIVIDERS (R_TOT ≈ 100 kΩ, V_SUPPLY = 4.94 V)
FORMULA:
  R2 = R_TOT × V_REF / 4.94 V
  R1 = R_TOT − R2
  (R1 to +5V, R2 to GND)

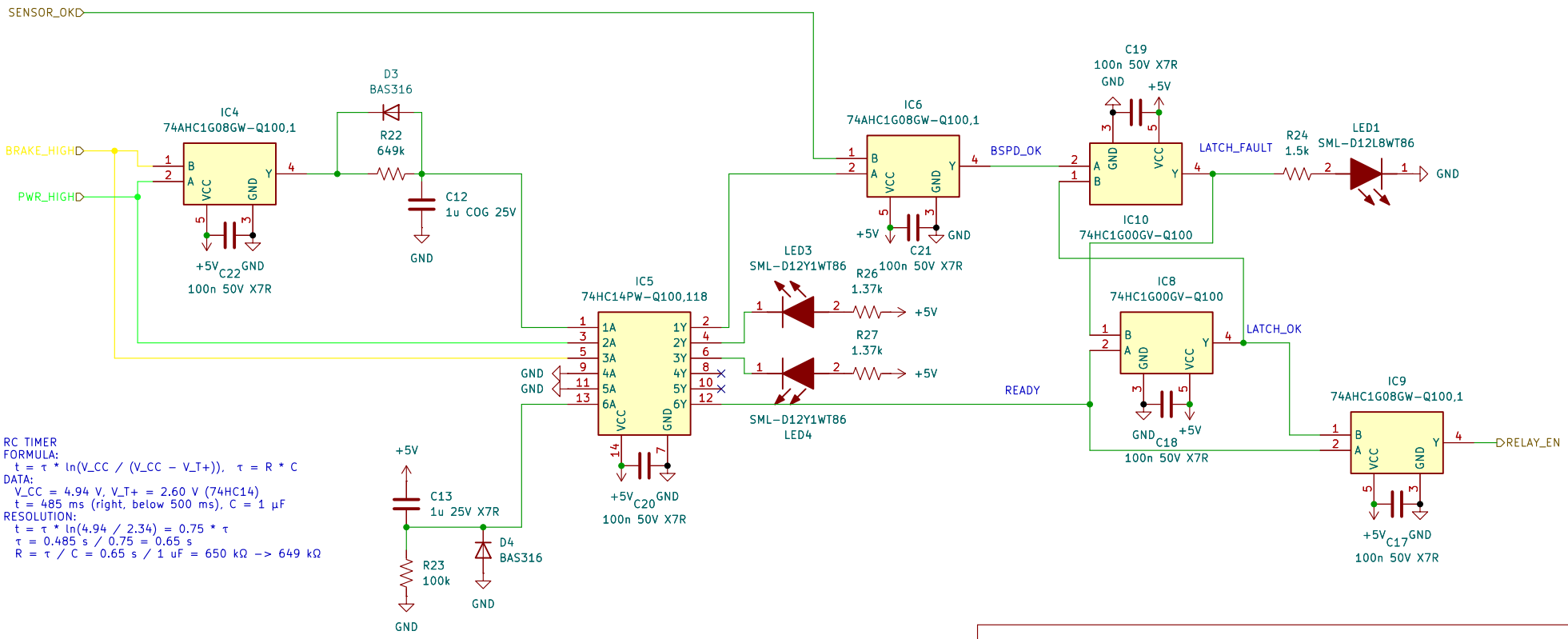
RESOLUTION:
  Brake threshold (V_REF = 1.683 V):
    R2 = 100 kΩ × 1.683 / 4.94 = 34.1 kΩ → 34.0 kΩ
    R1 = 100 kΩ − 34.1 kΩ = 65.9 kΩ → 66.5 kΩ
    Check: V_REF = 4.94 × 34.0 / 100.5 = 1.671 V

  Power threshold (V_REF = 2.640 V max):
    R2 = 100 kΩ × 2.640 / 4.94 = 53.4 kΩ
    R1 = 100 kΩ − 53.4 kΩ = 46.6 kΩ
    → E96 pair below 2.640 V: R1 = 44.2 kΩ, R2 = 49.9 kΩ
    Check: V_REF = 4.94 × 49.9 / 94.1 = 2.620 V

  Window (V_LOW = 0.25 V, V_HIGH = 4.75 V):
    R_WH = 100 kΩ × (4.94 − 4.75) / 4.94 = 3.85 kΩ → 3.83 kΩ
    R_WL = 100 kΩ × 0.25 / 4.94 = 5.06 kΩ → 5.11 kΩ
    R_WM = 100 kΩ − R_WH − R_WL = 91.1 kΩ → 90.9 kΩ
    Check: V_HIGH = 4.94 × 96.01 / 99.84 = 4.750 V
           V_LOW = 4.94 × 5.11 / 99.84 = 0.253 V

FORMULA: ΔV = V_IN * R_S / R_H
DATA: V_IN = 4,94V, RS = 10k, R_H = 1M
ΔV = 4,94V * 10k/100M = 0.049V = 49mV

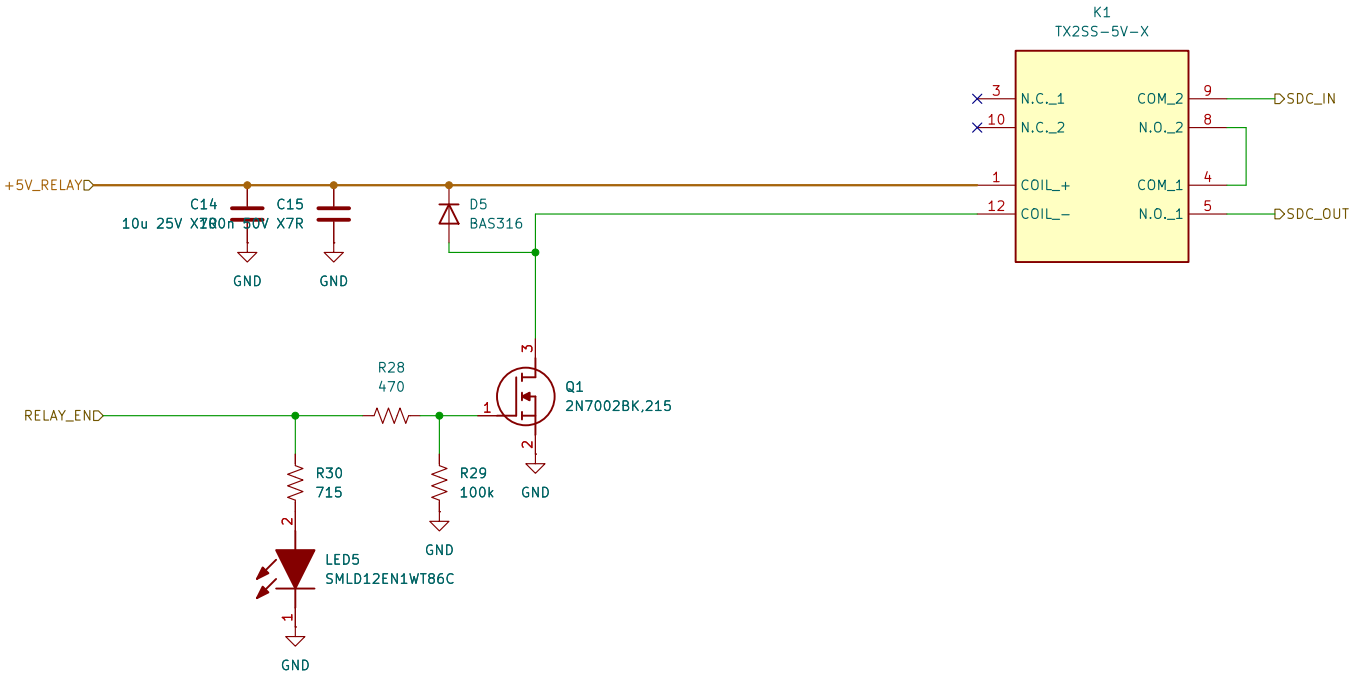
Sheet: Analog File: analog.kicad_sch		
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RC TIMER
FORMULA:
  t = τ * ln(V_CC / (V_CC − V_T+)), τ = R * C
DATA:
  V_CC = 4.94 V, V_T+ = 2.60 V (74HC14)
  t = 485 ms (right, below 500 ms), C = 1 µF
RESOLUTION:
  t = τ * ln(4.94 / 2.34) = 0.75 * τ
  τ = 0.485 s / 0.75 = 0.65 s
  R = τ / C = 0.65 s / 1 uF = 650 kΩ -> 649 kΩ

START UP RESET
FORMULA:
  t = τ * ln(V_CC / V_T−), τ = R * C
DATA:
  V_CC = 4.94 V, V_T− = 1.54 V (74HC14)
  t = 100 ms (10 times the settling time), R = 100 kΩ
RESOLUTION:
  t = τ * ln(4.94 / 1.54) = 1.17 × τ
  τ = 0.1 s / 1.17 = 0.086 s
  C = τ / R = 0.086 s / 100 kΩ = 0.86 uF -> 1uF

Sheet: Logic
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Sheet: Output
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